

GRAPHTHEORY-VL: VISUAL INSTANCES FOR EXACT MATHEMATICAL REASONING

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ABSTRACT

A mathematical drawing specifies an instance, while a solution must recover that instance and compute its requested target. A final answer alone can hide an incorrect structural interpretation, and removing the drawing can make a reasonable refusal indistinguishable from a failed solution. We introduce GraphTheory-VL, a collection of 62 visual mathematics problems with an embedded 20-problem hard subset, linking graph, algebraic, and topological targets to reference checks and complete model responses. We evaluate three model families with four responses per problem and compare original-image and text-only inputs on the hard subset. Mean accuracy on the 62-problem collection ranges from 10.89% to 31.85%, while the hard subset remains at 1.25% to 5.00%. Across models, at least one response succeeds on 43 of 62 problems, but only five of the 20 hard problems. An analysis of all 120 first responses in the image ablation distinguishes reasonable uncertainty, unsupported instance assumptions, and observable mathematical errors. In particular, a text-only response can match the target after guessing an omitted parameter, whereas a response that correctly identifies a graph can still compute the wrong critical group. These results connect repeated correctness to the mathematical and visual evidence needed to interpret it.

1 INTRODUCTION

A drawing can define a mathematical object without listing its structure in words. Curves determine adjacency, arrowheads determine orientation, and colored cells determine membership in a complex. The requested answer can be a spanning-tree count, a group invariant, or a symbolic polynomial. Solving the problem requires both the appropriate instance and the appropriate mathematics, which need not fail together.

Visual mathematics benchmarks evaluate diverse visual contexts, the distribution of information between text and images, and university-level mathematics (Lu et al., 2024; Zhang et al., 2025; Wang et al., 2026b). Graph reasoning benchmarks examine structural computation and its dependence on visual representation (Tang et al., 2025; Li et al., 2024; Sun et al., 2026). These resources motivate closer inspection of drawings that determine finite objects with exact targets. A plausible interpretation can support valid calculations for the wrong object, and a coarse invariant can agree when the requested richer invariant does not.

Consider the critical group of a graph. Its order equals a spanning-tree count, but does not identify the group (Biggs, 1999). One response studied here recognizes the correct graph and reports factors with the correct product, yet they define a different group. Conversely, without an image, a model can guess an unspecified parameter, calculate correctly conditional on that guess, and match the target. These responses separate a correct instance with wrong mathematics from a correct target without evidence for its instance assumption. Interpreting accuracy requires examining the supporting statements.

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GraphTheory-VL organizes this examination through 62 problems and an embedded 20-problem subset for detailed analysis. Each problem has an image, accepted target, content description, and reference-checking evidence. Data, code, and archived responses are available in the accompanying repository.¹ Targets span integers, tuples, finitely generated abelian groups, and multivariate polynomials. The collection includes direct counting and more involved algebraic or topological computations; its common focus is the relationship between a depicted instance and an exact mathematical target.

Challenge62 uses first-response failures of two screening models, while Hard20 uses a zero-observed-success screen across five models and four indexed attempts, followed by reference examination and quality filtering. The membership rule uses screening labels and reference-quality decisions, not independent-evaluation scores; some independent responses predate the final quality audit. Claude, the third evaluated family, did not participate in screening. We ask which selected failures recover in new attempts and which remain difficult. Established any-attempt and all-attempt success criteria (Chen et al., 2021; Yao et al., 2025) expose the distribution of these outcomes.

The three models reach mean accuracies of 31.85%, 28.63%, and 10.89% on Challenge62, but only 5.00%, 2.50%, and 1.25% on Hard20. Their combined success covers 43 problems, including five hard problems. Reading all 120 first responses on Hard20 under original-image and text-only inputs reveals reasonable refusals, unsupported instance assumptions, and explicit structural or mathematical conflicts. It supplies concrete counterexamples to treating target agreement as evidence of an instance-grounded solution.

The resource links 62 question–image pairs to accepted targets and reference evidence, 984 scored positions to their available answers and judgment sources, and all 120 hard first responses to instance and solution annotations. Repeated-response evaluation measures recovery after screening; the response analysis identifies why matching a coarse invariant, matching a complete target, and supporting a solution are different evaluation outcomes. This combination makes specific failures inspectable and supplies cases for testing visual extraction, symbolic calculation, and uncertainty handling.

2 RELATED WORK

Visual mathematics benchmarks differ in both their mathematical targets and the instance information supplied through each modality. MathVista and MATH-Vision evaluate diverse visual mathematical problems, with the latter drawing on competition questions (Lu et al., 2024; Wang et al., 2024). MathVerse varies the distribution of information between text and diagrams across six versions and evaluates intermediate reasoning steps (Zhang et al., 2025). MathSight extends this line to university mathematics with original, hand-drawn, photographed, and image-free conditions (Wang et al., 2026b). Their public assets include problems, images, and reference answers or solutions. Removing an image need not provide a replacement description; our image-free condition preserves the original text. VCBench emphasizes explicit visual dependencies (Wang et al., 2025), while DynaMath and MathCheck examine variation and generalization across related mathematical tasks (Zou et al., 2025; Zhou et al., 2025). These studies establish input conditions and task variants as central dimensions of evaluation.

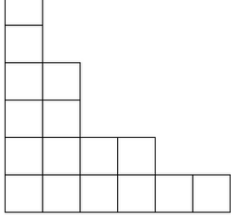
Graph benchmarks provide a more direct comparison of mathematical objects. GraphArena evaluates graph computation (Tang et al., 2025). VisionGraph and GITA connect graphical representations with multistep reasoning (Li et al., 2024; Wei et al., 2024). Visual Graph Arena tests reasoning across different layouts, including graph isomorphism, while VGCure examines fundamental graph understanding and reasoning skills (Babaiee et al., 2025; Zhu et al., 2025). CombiGraph-Vis supplies discrete olympiad problems, solutions, technique labels, and repeated-response evaluation (Mahdavi et al., 2025). Its inspected image-bearing examples include jump counting, grid coverage, constrained coloring, and tiling. GraphVerse specifies eleven tasks over single or paired graph images, including shortest paths, coloring, graph edit distance, and maximum common subgraphs, with algorithmic reference answers and process-sensitive scoring (Sun et al., 2026). Its task list provides a concrete comparison with our critical-group, homology, and multivariate determinant tar-

¹<https://anonymous.4open.science/r/GraphTheory-VL-Experiments-2027/>

Hard20 $\subset$ Challenge62 $\subset$ source pool (623)

(a) Representation

GT-0004



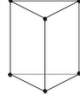
$$\lambda = (6, 4, 2, 2, 1, 1) = \lambda'$$

Target: constituent dimension;
absolute character difference.

$$(d, \rho) = (3^6 \cdot 7^2 \cdot 13, \sqrt{55})$$

(b) Critical groups

GT-0411



$$G = K_{2,2,2,2,2}$$

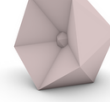
H : triangular prism

Target: nonunit Smith factors.

$$((3, 12, 48, 48), (5, 15))$$

(c) Cell symmetries

GT-0514



$$n = 8 \text{ sectors}; |G| = 16$$

Target: $(16, Z_G, 1)$

$$16Z_G = a_1^8 + 5a_2^4 + 2a_4^2 + 4a_8 + 4a_1^2 a_2^3$$

Figure 1: Visual instances connect diagrams to exact targets. Original source images are displayed beside mathematical descriptions; their targets include character expressions, critical groups, and a symmetry cycle index. Challenge62 contains Hard20. Membership uses screening labels and reference-quality decisions; independent responses are scored on these retained sets.

gets. These examples motivate comparing exact requested outputs and available verification assets, beyond broad labels such as graph reasoning or university mathematics.

GraphTheory-VL organizes its contribution around visual instances whose exact mathematical targets can be recomputed. For example, adjacency determines the critical group, whose invariant factors can be obtained by Smith normal form (Biggs, 1999; Stanley, 2016), and embedded incidence data determines the relevant homology calculations (Hatcher, 2002). Each content card connects these instance facts to an accepted target and its verification evidence. This also supports analysis beyond whether the final answer matches. We-Math studies component knowledge through decomposed questions (Qiao et al., 2025), and process supervision distinguishes feedback on intermediate steps from feedback on outcomes (Lightman et al., 2024). Description-based interventions have also examined perception and inductive reasoning in abstract reasoning tasks (Wang et al., 2026a). Our study instead provides an observational analysis of complete responses against explicit instance facts and exact mathematical targets. It identifies conflicts that a final score can hide; it does not estimate the causal contributions of perception and reasoning.

3 BENCHMARK CONSTRUCTION

3.1 VISUAL INSTANCES AND EXACT TARGETS

GraphTheory-VL pairs a mathematical question with a diagram and an accepted target. Incidence, directions, multiplicities, colors, or identifications specify a finite object, while the question defines the requested computation. An instance is $(t_i, I_i, \mathcal{Y}_i)$: question, image, and acceptable mathematical answers. Equivalent expressions can represent the same target, but matching one component of a requested tuple does not satisfy the complete question.

The targets expose distinct mathematical requirements. GT-0411 asks for critical groups from the Smith factors of two reduced Laplacians. GT-0019 converts strand endpoints into a permutation, a hypermap genus, and a spectral minimal polynomial requiring an irreducibility argument. GT-0560 converts signed, labeled arrows into a matrix over $\mathbb{Z}[x, y]$ and asks for the expanded $\det(I - zM)$. Each content card connects the mathematical object, visual facts, complete target, and reference evidence. The 62 outputs comprise 19 integer tuples, 14 symbolic polynomials, 10 symbolic tuples, eight integers, seven group-valued outputs, and four structured lists.

3.2 SELECTION AND DATASET MEMBERSHIP

We construct *Challenge62* and its nested *Hard20* from 623 question–image pairs. Screening uses four indexed positions from GPT-5.6 Luna, Gemini 3.6 Flash, DeepSeek V4 Flash Vision, Doubao Seed 2.1 Pro, and Qwen3-VL-32B-Instruct. Challenge requires both GPT and Gemini to receive `Incorrect` on their first screening response. Hard requires zero observed `Correct` labels across all five models’ four positions. These conditions use archived screening answers evaluated against corrected references, followed by reference-quality exclusions. GT-0050 has a missing third DeepSeek screening response: Hard therefore denotes no observed success, not twenty completed failures per question.

Independent response scores are not an input to the membership rule, and later success does not remove a question. Some such responses were available during reference-quality work; that work was neither preregistered nor documented as blinded (Appendix A.9). Challenge supports the common model comparison, while Hard supports full first-response analysis. Difficulty comparisons use disjoint Hard20 and the remaining 42 Challenge questions. These selected instances do not form a random sample of visual mathematics.

3.3 REFERENCE EXAMINATION AND RETAINED EVIDENCE

Reference examination combines model-assisted review, AI-assistant inspection of images and questions, mathematical derivation, and exact computation. The 134-question audit and two earlier corrections yield 80 corrected references in total (Appendix A.2). Affected archived answers are rejudged before reconstructing membership, with missing responses retained as missing. Corrected-reference screening gives 83 Challenge and 28 Hard candidates. Excluding 16 unresolved construction or reference problems and five awaiting complete verification leaves 62 and 20. Exclusion decisions arose during the audit and were not preregistered.

All 62 retained questions have explanations and traceable reference evidence: 35 retain supported targets and 27 have corrected targets. The companion binds each question and image to its target, supporting structure claims, and available calculations; its index distinguishes checker source from recorded execution (Appendix A.10). Exact arithmetic checks a reconstructed object, while agreement between that object and the drawing remains a separate verification step. Neither step has dataset-wide independent human acceptance.

The 62 retained questions and their diagrams were created by our team. Each question–image pair maps to a unique source-workbook row, preserving its connection to the experimental records and reference evidence. GT-0560’s directional wording error is documented without replacing its evaluated input (Appendix A.7).

A separate text-sufficiency audit identifies an unspecified instance fact for every Hard20 target. Some components and general formulas still follow from text. Appendix H records the missing facts and alternative-instance arguments, including their verification limits. Challenge also contains text-sufficient cases, including GT-0573; the Hard finding does not extend automatically to the collection.

4 EVALUATION PROTOCOL

Models and inputs. We evaluate GPT-5.6 Luna, Gemini 3.6 Flash, and Claude Opus 4.8 with four separate responses per question and condition. Claude did not participate in screening. Independent evaluation uses answer records separate from screening, reusing existing independent answers to identical inputs. This separation does not assert independence of training data or stochastic outputs. Requests provide the question and any options, but no reference, solution, prior response, or screening label. Solvers are instructed to avoid external tools and return `answer` and `reasoning` fields.

All models receive a 128,000-token output ceiling. GPT and Gemini request high reasoning effort, whereas Claude leaves effort unspecified. Sampling otherwise uses provider defaults. The task-order seed 20260823 is not an API sampling seed. Equal ceilings do not imply equal realized computation, and service identifiers do not independently authenticate the underlying model weights.

Appendix A.3 documents the identifiers and recovery through Krill, BUZZ, and OpenLux. The original-image condition O supplies the question and image. GPT and Gemini transmit original bytes, while Claude uses the original image or lossless PNG preparation. T removes the image while preserving all wording, including the instruction referring to a supplied image. Its behavior is measured under that retained prompt.

Challenge62 is evaluated in O and Hard20 additionally in T, giving $62 \times 3 \times 4 + 20 \times 3 \times 4 = 984$ unique scoring positions. Hard O is included in Challenge O. There are 983 complete archived responses and one user-reported `Incorrect` decision for Claude T, GT-0560, response 2, whose inspected full answer is unavailable locally. Appendix A.5 reports the complete-archive sensitivity. Technical recovery fills missing positions without replacing completed answers based on correctness.

Answer evaluation. Correctness requires the complete requested target under the current reference, allowing established equivalence. Explicit proof or certificate requirements belong to that target. Otherwise, intermediate reasoning is assessed separately. T retains the original-target endpoint, so justified abstention can be appropriate behavior without a target hit. Labels comprise 832 DeepSeek judgments, 151 strict or deterministic comparisons, and one user decision, totaling 181 `Correct` and 803 `Incorrect`. Additional explicit-target checks cover all 181 correct responses and agree with their labels. They check the positive-label targets, not the false-negative rate among the 803 incorrect labels, and do not supply independent human calibration (Appendix A.4).

Metrics and uncertainty. For N questions, let $y_{ir} \in \{0, 1\}$ indicate correctness of indexed response $r \in \{1, 2, 3, 4\}$ to question i , and $c_i = \sum_{r=1}^4 y_{ir}$. We report

$$\begin{aligned} \text{Acc@1} &= \frac{1}{N} \sum_i y_{i1}, & \text{MeanAcc} &= \frac{1}{4N} \sum_i c_i, \\ \text{Pass@4} &= \frac{1}{N} \sum_i \mathbf{1}[c_i > 0], & \text{Stable@4} &= \frac{1}{N} \sum_i \mathbf{1}[c_i = 4]. \end{aligned} \tag{1}$$

These established at-least-one and all-attempt criteria (Chen et al., 2021; Yao et al., 2025) use the same four responses without answer aggregation. Acc@1 fixes the first response. The distribution of c_i distinguishes occasional from repeated success.

MeanAcc intervals use 100,000 percentile bootstrap replicates over questions (Efron, 1979), keeping four responses together and resampling common questions jointly for comparisons. Binary question-level metrics receive Wilson intervals (Wilson, 1927). The bootstrap seed is 20260910. Comparisons are exploratory, unadjusted for multiplicity, and conditional on this selected benchmark. Appendix A.5 details all-zero groups and paired populations.

5 INDEPENDENT EVALUATION

Screened failures can recover. Table 1 reports complete coverage of Challenge62 for all three models. GPT achieves 79 correct responses out of 248, Gemini 71, and Claude 27. Their first-response accuracies are 21/62, 17/62, and 6/62, respectively. These are new responses: a failure in the screening run does not imply failure on the first independent attempt. The mean accuracies of 31.85%, 28.63%, and 10.89% also show that Challenge62 is not uniformly at the accuracy floor.

Repeated attempts expand observed success, but they seldom produce success on every attempt. GPT solves 34 problems at least once and all four times on eight; Gemini reaches 27 and nine; Claude reaches 14 and one. Relative to each model’s first independent response, the remaining three responses recover 13, 10, and eight additional problems. These counts measure recovery among the fixed four responses, without introducing an answer-selection procedure.

Hard20 retains a distinct difficulty profile. On Hard20, the models produce only four, two, and one correct original-image responses out of 80. GPT succeeds on three problems, Gemini on one, and Claude on one; no model answers any hard problem correctly on all four attempts. Figure 2 shows this concentration directly rather than reducing it to a single average.

Table 1: Independent original-image evaluation. Rates and their intervals are percentages. Each model contributes four responses per problem. MeanAcc averages all four responses; Pass@4 and Stable@4 measure at least one and four correct responses. Intervals are 95% intervals computed at the problem level. Hard20 is contained in Challenge62.

Subset	Model	Correct	MeanAcc	Pass@4	Stable@4
Challenge62	GPT-5.6 Luna	79/248	31.85 [23.0, 41.1]	54.84 [42.5, 66.6]	12.90 [6.7, 23.4]
	Gemini 3.6 Flash	71/248	28.63 [19.4, 38.3]	43.55 [31.9, 55.9]	14.52 [7.8, 25.3]
	Claude Opus 4.8	27/248	10.89 [5.6, 16.9]	22.58 [14.0, 34.4]	1.61 [0.3, 8.6]
Hard20	GPT-5.6 Luna	4/80	5.00 [0.0, 11.2]	15.00 [5.2, 36.0]	0.00 [0.0, 16.1]
	Gemini 3.6 Flash	2/80	2.50 [0.0, 7.5]	5.00 [0.9, 23.6]	0.00 [0.0, 16.1]
	Claude Opus 4.8	1/80	1.25 [0.0, 3.8]	5.00 [0.9, 23.6]	0.00 [0.0, 16.1]

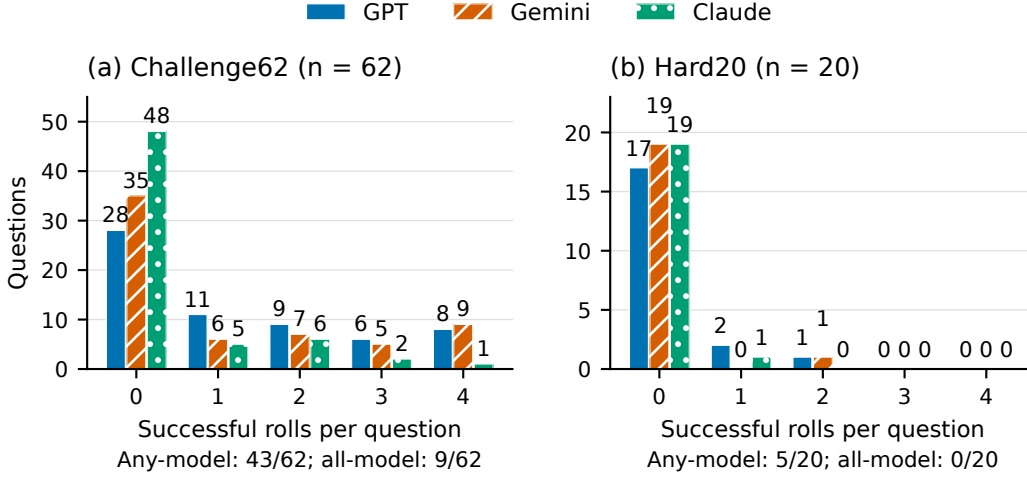


Figure 2: Distribution of the number of correct responses per problem. The same four responses underlie all accuracy and repeated-success metrics. Challenge62 contains both recoverable and persistent failures; the distribution on Hard20 is concentrated at zero successes.

The disjoint comparison with the other 42 Challenge problems is informative. On that group, mean accuracy is 75/168 (44.64%) for GPT, 69/168 (41.07%) for Gemini, and 26/168 (15.48%) for Claude. At least one model succeeds on 38 of these 42 problems, compared with five of the 20 hard problems. Thus, the broader collection offers opportunities to observe recovery, while the nested subset concentrates persistent failures. This contrast describes the sets produced by the screening procedure; it is not an estimate of the difficulty distribution of visual mathematics in general.

Success sets overlap only partly. Across the three models and four responses, 43 of the 62 problems have at least one successful response. Nine problems are solved at least once by all three models. Of the 19 problems with no observed success, 15 are in Hard20 and four are outside it. The union is an observational coverage statistic: selecting the correct response from this pool would require an additional decision rule that is not evaluated here.

The paired mean-accuracy difference between GPT and Gemini on Challenge62 is 3.23 percentage points, with a 95% interval of $[-6.85, 13.31]$. The GPT–Claude and Gemini–Claude differences are 20.97 points $[11.29, 30.65]$ and 17.74 points $[10.48, 25.81]$, respectively. These exploratory compar-

Table 2: Hard20 image ablation. Text-only inputs remove the image while retaining the problem text and original instruction. Rates are percentages. Scores measure agreement with the original target. The final column compares the two conditions on the same 20 problems; all differences are in percentage points.

Model	O correct	T correct	O MeanAcc	T MeanAcc	Δ_{T-O} [95% CI]
GPT-5.6 Luna	4/80	0/80	5.00	0.00	-5.00 [-11.25, 0.00]
Gemini 3.6 Flash	2/80	0/80	2.50	0.00	-2.50 [-7.50, 0.00]
Claude Opus 4.8	1/80	4/80	1.25	5.00	+3.75 [-3.75, 15.00]

isons refer to the selected problems and recorded service configurations. They do not isolate model architecture or establish a general ranking.

Removing the image changes what accuracy means. On Hard20, GPT and Gemini produce no correct text-only targets in 80 responses each. Claude produces four, all on one problem. The text-only minus original-image differences are -5.00 , -2.50 , and $+3.75$ percentage points. Their paired intervals all include zero (Table 2). More importantly, an incorrect text-only target may accompany a mathematically justified statement that the instance is unspecified. We therefore interpret this ablation together with the complete first-response analysis in Section 6.

Removing GT-0560, whose directional wording conflicts with its drawing, gives the same 19-question population for all models. Original-image MeanAcc becomes 5.26%, 2.63%, and 0%; Claude retains its four T hits on GT-0514. The pattern of sparse Hard success therefore persists. Appendix G gives all metrics and paired intervals; this exclusion does not replace evaluation with corrected wording.

6 INSTANCE AND SOLUTION ANALYSIS

We inspect all 120 first responses on Hard20, with 20 per model-condition cell, avoiding selection of unusually favorable or unfavorable attempts. Explicit statements are checked against the image, current reference, and mathematical evidence. Model-assisted review precedes an AI assistant’s full-text adjudication with references and prior labels visible. These are model-assisted annotations; Appendix A.6 describes their scope and uncertainty labels.

6.1 WHAT REMAINS DETERMINED WITHOUT THE IMAGE?

The text-sufficiency audit identifies missing instance facts for all 20 complete Hard targets. Appendix H makes the per-question arguments inspectable: it lists the omitted fact, a permitted variation, and the affected target component. Most are concrete substitutions or local graph changes; a few retain qualitative construction arguments and are flagged as such. In a permutation problem, for example, the abstract text permits both a cyclic permutation and its inverse, but these choices yield different genera. In a graph problem, adding a loop changes the cycle rank without changing the spanning-tree count. These variations preserve the relevant textual constraints while changing a requested component.

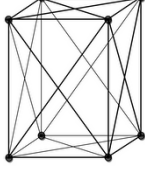
Underdetermination of the complete target does not mean that nothing can be derived. In the sector-complex problem GT-0514, the text specifies a central polygon with attached triangles. It determines the Euler characteristic and a family of dihedral actions, but not the number of sectors. A text-only GPT response correctly keeps that number symbolic and derives the corresponding general expressions. Its failure to match the fixed numerical target is therefore different from an invalid derivation. The finding applies to Hard20; the broader collection also contains text-sufficient cases.

6.2 TEXT-ONLY RESPONSES EXPRESS DIFFERENT KINDS OF UNCERTAINTY

Table 3 separates five text-only answering strategies. Of the 60 responses, 24 reasonably decline to determine the target and four provide a general method or formula. Nine retain an explicit guess in their final answer, two incorrectly interpret the missing image as an empty object, and 21 commit

(a) Structure correct, factors wrong

GT-0411 · Gemini · original image · roll 1

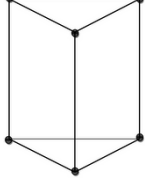


Identified $K_{2,2,2,2}$ and the triangular prism.

Upper-graph factors:
Observed: (6, 6, 48, 48)

Exact: (3, 12, 48, 48)

Both products are 82944.

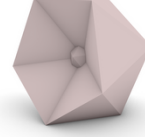


$\dim_{\mathbb{F}_2} K(G)/2K(G) = 3$,
versus 4 for the reported factors.

Gcds of all minors give the Smith diagonal (1,1,1,3,12,48,48).

(b) Target hit, absent evidence

GT-0514 · Claude · text only · roll 1



Original image (withheld in T)

“The rendering shows $n = 8$ sectors.”

The guessed sector count matches.
The complete target passes; the claim of visual evidence is unsupported.

Text fixes $\chi(K) = 1$ and formulas in n .

It does not fix n or the full triple.

Figure 3: Two ways a final target and its supporting evidence can diverge. In GT-0411, the correct graph identity precedes incorrect group factors. In GT-0514, a text-only answer guesses the true sector count and then attributes it to an absent rendering. Original images define the benchmark instances; the text-only response did not receive its image.

Table 3: Primary categories for all first responses on Hard20. Each condition contains 60 responses, with 20 per model. Text-only categories follow the final answering stance; original-image categories identify one directly supported conflict or outcome per response. Visual attribution is a separate, potentially overlapping dimension.

Category	GPT /20	Gemini /20	Claude /20	Total /60
<i>O: observed primary category (one per response)</i>				
Instance statement conflict	16	17	18	51
Modeling conflict	0	1	1	2
Condition conflict	0	0	1	1
Computation / logic conflict	0	1	0	1
Error stage unresolved	2	0	0	2
Target and checked reasoning consistent	2	1	0	3
<i>T: final response strategy (one per response)</i>				
Justified refusal	17	0	7	24
General formula / method	1	3	0	4
Final answer remains a guess	0	0	9	9
Missing image treated as empty	0	0	2	2
Unsupported instance answer	2	17	2	21
<i>T: attribution to the absent image (separate dimension)</i>				
Explicit attribution	1	5	2	8
Ambiguous attribution	1	4	1	6

to instance facts unsupported by the supplied text. The distinction between the last two categories matters: absence of an image gives no basis for setting a vertex set or graph to the empty set.

GPT accounts for 17 reasonable refusals, Gemini for 17 unsupported instance assignments, and Claude for all nine final guesses. These frequencies describe the sampled responses under the recorded configurations, rather than stable traits of a model family.

A separate annotation asks whether the response explicitly attributes a fact to an image that it did not receive. Eight of 60 responses meet this criterion, while six remain ambiguous. These observations are conditional on the retained instruction referring to a supplied image. A numerical assertion alone does not establish visual attribution; unsupported assumptions and claims of visual evidence remain separate categories.

6.3 TARGET AGREEMENT CAN CONCEAL AN UNSUPPORTED INSTANCE

GT-0514 provides a concrete example. The true instance has eight sectors. Claude’s first text-only response initially guesses this number and later states, “The rendering shows $n = 8$ sectors.” The resulting group order, cycle index, and Euler characteristic match the accepted target, and its conditional algebra is valid. Nevertheless, the quoted observation cannot follow from the supplied input, which contains no rendering. All four Claude text-only target hits occur on this problem.

GPT preserves the unresolved parameter and misses the fixed target; Claude commits to its true value without visual evidence and matches it. Reporting the target score alongside the answering behavior preserves this difference under the same scoring rule.

6.4 A CORRECT INSTANCE CAN STILL LEAD TO A WRONG EXACT TARGET

Among the 60 original-image first responses, 51 contain an explicit instance statement that conflicts with the image evidence or verified structure. Two contain modeling conflicts, one violates a mathematical condition, and one has a computational conflict after identifying the relevant graph. Two errors cannot be localized from the response, and three responses agree with the target and the derivation examined. These labels describe observable output, not internal perception failures. GT-0560 also has contradictory wording; excluding all six of its first responses leaves 49 instance-conflict annotations among 57 O responses (Appendix G). The conflicting input prevents attributing those removed discrepancies solely to a model.

In GT-0411, Gemini correctly identifies the upper graph as $K_{2,2,2,2}$, but gives nontrivial critical-group factors $(6, 6, 48, 48)$ instead of $(3, 12, 48, 48)$. Both lists satisfy the divisibility convention and have product 82944. Yet their quotients modulo 2 have dimensions four and three, respectively, so they cannot describe isomorphic groups. Determinantal-divisor checks on the reduced Laplacian confirm the latter factors (Appendix B). The failure occurs beyond identifying the graph and cannot be detected by checking its spanning-tree count alone.

We further inspect all three first responses in four diagnostic cases: GT-0411, GT-0382, and GT-0019 in O, and GT-0514 in T. Among the nine O responses, four match the chosen coarse or terminal-value projection, but only one satisfies its corresponding complete target. GT-0382’s missed loop preserves the spanning-tree count; GT-0019’s proof obligation rejects a wrong derivation with matching terminal values. This post-hoc case census (Appendix B.5) demonstrates the diagnostic distinction, not its prevalence across the dataset.

7 DISCUSSION AND CONCLUSION

GraphTheory-VL connects visual instances, exact targets, and repeated responses. Independent evaluation separates recoverable Challenge failures from persistent Hard failures. Complete responses distinguish valid calculations on wrong instances from target hits without supporting evidence.

The resource supports concrete investigations of visual extraction and symbolic computation. A solver can expose the graph, permutation, or cell structure it inferred, allowing comparison with the reference before the subsequent calculation is judged. A parameterized answer can retain uncertainty when the input omits an instance fact. The present measurements provide examples and reference computations for such investigations; they do not measure improvements from a new solving method.

The findings depend on model-conditioned selection, a small hard subset, and third-party service configurations. Reference and behavior checks use AI assistance and exact computation without independent human calibration; positive-label checks do not estimate missed correct answers. Reference corrections substantially affect fixed-response scores. The companion enables inspection of

these decisions. Structural-input interventions and later-attempt behavior remain unmeasured. The results support evaluating the visual and mathematical statements behind a target alongside repeated correctness.

AI USE STATEMENT

Generative AI tools were used in developing the research framing and experimental design, implementing evaluation and analysis code, checking and revising mathematical references, formulating and checking mathematical arguments, organizing the data, analyzing responses, interpreting results, and drafting this manuscript. DeepSeek supplied semantic answer judgments; Claude supplied preliminary qualitative assessments; Codex assisted with data integration, exact-computation checks, evidence examination, analysis, and writing. Human review is recorded only where it was explicitly supplied. Reference and response checks combine source-linked inspection with exact computation, as described in Sections 3 and 4 and Appendix A. These procedures are not represented as independent human verification of the full collection.

REPRODUCIBILITY STATEMENT

The evaluation is specified by fixed problem memberships, reference targets, input conditions, model identifiers, and four indexed responses per problem. Appendix A describes the scoring and statistical procedures, and Appendices C and D give per-problem outcomes and the content index. Data, experimental code, and recorded responses are accessible through the accompanying repository linked in Section 1. The artifact includes the current question–image pairs, full available responses, label sources, reference evidence, and the earlier screening and evaluation records supporting selection and correction. Its companion index supports per-question inspection (Appendix A.10). Credential-free code, exact prompts, and configurations support rerunning the procedures with separately supplied service access. The archive preserves the distinction between 983 complete current responses and one user-adjudicated position, and between checker source and recorded execution. For v1.0, we use CC BY 4.0 for the 62 team-created questions and diagrams and the MIT License for team-authored evaluation code. External materials retain their respective terms.

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A ADDITIONAL CONSTRUCTION AND EVALUATION DETAILS

A.1 SCREENING, QUALITY DECISIONS, AND FIXED MEMBERSHIP

The source pool contains 623 identified questions, their images, reference answers, and archived screening records. The five requested screening identifiers are `deepseek-v4-flash-vision-exp`, `doubao-seed-2-1-pro-260628`, `gemini-3.6-flash`, `gpt-5.6-luna`, and `qwen3-v1-32b-instruct`. Four scheduled positions per model produce 12,460 positions. With the active reference corrections, these contain 5,107 correct, 7,345 incorrect, and eight missing labels. Screening records and independent evaluation records are maintained separately.

Before quality exclusions, the two first-response failures identify 83 Challenge candidates and the zero-observed-success condition identifies 28 Hard candidates. The 16 questions with unresolved problems and five awaiting reference verification are excluded from evaluation membership. All 21 belong to Challenge, and eight also belong to Hard. The retained Hard20 screening records contain 399 incorrect positions and one missing position, GT-0050’s third DeepSeek response, which remains missing.

The five references awaiting verification are GT-0322, GT-0345, GT-0347, GT-0349, and GT-0598. Their exclusion concerns incomplete evidence rather than a confirmed wrong answer. Other exclusions have question–image conflicts or refuted targets without resolved replacements. Membership follows these audit decisions and remains fixed when reporting independent evaluation responses.

A.2 REFERENCE CORRECTIONS AND MATHEMATICAL EVIDENCE

The fixed 134-question audit contains 78 reference corrections, 35 supported references, 16 unresolved problems, and five incomplete verifications. Two additional corrections, GT-0148 and GT-0625, were recorded separately, giving 80 in total. The audit combines model responses with direct image examination, mathematical derivation, and executable checks by an AI assistant. Among retained Challenge questions, 35 original targets are supported and 27 have been corrected. Every retained question has an explanation and a location for the mathematical evidence used to support its current target.

Evidence takes a form appropriate to the object. Exact arithmetic checks matrix invariants, integer products, and coefficients of identical polynomial monomials. Invariant-factor questions require group structure, not merely group order. Image interpretation remains separate: exact computation on an incorrect adjacency matrix would not validate the source instance. The reference index links structure claims to images or question rules and computations to their outputs.

The 78 corrections prompted 27 rejudgment batches covering 2,671 complete archived answers: 1,559 screening and 1,112 independent answers. Strict offline comparisons resolved 1,233, and DeepSeek evaluated the other 1,438. The labels contain 633 correct and 2,038 incorrect answers, with 775 changes, including 255 independent-evaluation changes. The two earlier corrections add 40 screening rejudgments. These purposively selected records do not estimate the error rate of the full source collection.

Reference sensitivity uses fixed question memberships and identical answer texts before and after correction and rejudgment. Neither new answers nor changes in membership enter that comparison. Its interpretation includes both reference changes and the application of a new judgment to an existing response. It is consequently separate from the independent Challenge62 and Hard20 results.

A.3 GENERATION PROMPTS, SERVICES, AND TECHNICAL RECOVERY

The exact common system message is:

You are solving an expert-level graph theory problem. Use both the problem text and the supplied image. Work independently and do not use external tools or external information. Return the final answer in the required format.

The user message begins with `Problem:`, a newline, and the complete question. Nonempty options are appended after a blank line as `Options:`, a newline, and their contents. After a blank line, every message ends with:

```
Return your response using exactly these two fields:
answer: <final answer>
reasoning: <concise explanation>
```

The companion contains the exact expanded messages for all 62 questions, verified against the stored prompt identifiers of all 983 complete responses. T removes the image block without changing either textual message. Statements about an absent image are therefore observed under an instruction that still refers to a supplied image.

The independent identifiers are `gpt-5.6-luna`, `gemini-3.6-flash`, and `claude-opus-4-8`. GPT uses `max_output_tokens=128000`, while Gemini and Claude use `max_tokens=128000`. GPT and Gemini request high reasoning effort, whereas Claude leaves effort unspecified. Temperature and nucleus-sampling settings use provider defaults. Records retain requests, responses, finish status, and available usage information. The task-order seed 20260823 is not transmitted as an API sampling seed.

Krill serves GPT and Claude requests, BUZZ serves the initial Gemini requests, and OpenLux serves Gemini recovery and further coverage as well as GPT recovery. The final 280-answer completion of the common Challenge table comprises 19 GPT responses through Krill, 121 GPT responses through OpenLux, and 140 Gemini responses through OpenLux. The same requested model, prompt, image, and output ceiling are retained when changing the service for these positions. Service changes are recorded alongside model identifiers because the providers' underlying routing and weights are not independently verified.

Technical failures and empty responses are recovered as missing positions, retaining failed attempts separately. A completed response is not replaced because it is incorrect. Challenge O contains 744 complete answers, comprising 464 reused independent answers and 280 additional answers, while Hard T contributes 239 complete answers and one manual label. Their 984 unique scoring positions count Hard O once, within Challenge O.

A.4 JUDGMENT SEMANTICS AND ADDITIONAL CHECKS

DeepSeek V4 Flash is the semantic judge, accessed under `deepseek-v4-flash`. The corrected-reference and completion prompts supply the current question, target and explanation, and full candidate response, without the previous correctness label. They request a target verdict, a separate reasoning status, and a mathematical rationale. For GT-0019, the target requires a valid irreducibility argument. For a question requesting a polynomial alone, unrelated intermediate errors are recorded separately from target equivalence.

Strict comparisons require an explicit complete target in a supported form, including ordered integer tuples and restricted exact integer expressions. Unclear extractions or unestablished equivalences are not automatically marked correct. The registry binds the 983 automatically scored positions to their question, current reference, full response, and judgment source; the manual exception binds to the user's decision record. Methods total 832 semantic judgments, 151 strict or deterministic comparisons, and one manual decision, with 344 positions covered by corrected-reference rejudgment.

Additional checks cover all 181 current correct responses, with no final-target disagreement. They compare explicit answers with visible references, using exact calculations where applicable, without independently validating every intermediate statement. This positive-only coverage cannot estimate the false-negative rate among 803 incorrect labels or exclude shared errors in references and judgments. A blinded, independently adjudicated sample of both labels has not been collected. An earlier audit of 396 responses received 394 model reviews and raised 33 issues. Thirty-two were adjudicated and one belongs to an excluded question. The audit overlaps 147 current positions, of which 62 retain the same reference. This is coverage information, not human calibration of judge accuracy.

A.5 STATISTICAL UNITS AND QUESTION EXCLUSION

Each question contributes four binary outcomes per model and condition. Bootstrap samples draw questions with replacement, carry their four outcomes together, and use the same indices on both sides of a comparison. The 95% percentile intervals use 100,000 replicates and linear interpolation of quantiles. Binary question-level metrics use Wilson intervals. Pass@4 and Stable@4 measure observed any-attempt and all-attempt success without selecting an answer for deployment.

All-zero groups produce a degenerate percentile interval for MeanAcc because every empirical re-sample remains zero. This does not establish a zero underlying success probability. With 20 questions and no observed success, the Wilson upper endpoint for Pass@4 is 16.11%. Model and O/T differences are exploratory and have no multiplicity correction. All intervals describe variation across this selected set rather than uncertainty from a random sample of all visual mathematics.

The principal paired Hard comparisons include the user-reported label described below. Appendix G removes GT-0560 from every model and condition, addressing both its incomplete archive and its conflicting wording on a common population. Excluding the whole question preserves four responses per condition; no answer or correctness label is changed.

A.6 FULL-RESPONSE BEHAVIORAL CODING

Behavioral analysis fixes response 1 for every Hard question, model, and condition before inspecting behavior, yielding 120 complete answers. Twenty question packs contain the image, text, current reference, mathematical evidence, and six responses. The Claude initial review sees randomized candidate letters and O/T conditions with model identity and original judgment hidden. Twenty reviews complete with a 16,000-token ceiling. Final AI-assistant adjudication sees the complete answers, references, original labels, and review suggestions, and is not blinded.

Text sufficiency is assessed per question by identifying an unspecified instance fact or compatible instances with different targets. A reference to a figure alone is insufficient evidence of underdetermination. Response coding distinguishes target hits, the stance of the final answer, specific conflicts with instance or mathematical facts, and explicit attribution of facts to an unprovided image. The summary assigns one primary O conflict and one primary T strategy per answer, while visual attribution is a separate dimension. Each positive claim is supported by a located quotation and corresponding source or mathematical evidence. The 134 indexed quotations have been matched to their full responses. Claude’s involvement in both solving and initial review, and the reference exposure of the final assistant checks, make these model-assisted annotations rather than independent human judgments. Quotation agreement establishes traceability, not the accuracy of a category assignment. No blinded independent recoding or inter-annotator agreement estimate is available; the category frequencies remain conditional on this protocol.

A.7 GT-0560 WORDING AND MANUAL JUDGMENT

GT-0560 describes two curved arrows as issuing from vertex 4. The drawing instead contains $4 \rightarrow 2$ and $3 \rightarrow 4$, whose entries follow the explicit tail-to-arrowhead convention. The reference uses these displayed directions. The evaluated wording is preserved and the inaccurate phrase documented as an erratum. This is a defect in the supplied item: a response compatible with the phrase cannot be treated as an unambiguous image-reading error. None of the three O first responses explicitly says that this phrase determined its chosen direction. Gemini’s relevant matrix entries are compatible with it, but the causal basis of those entries cannot be inferred from the answer. The uniform exclusion in Appendix G therefore complements the main scores without assigning the discrepancy solely to the model.

Claude’s second T response for this question has three incomplete local stream records. The user inspected an answer and explicitly judged it incorrect, but that complete inspected answer was not supplied to the local archive. We retain the decision as one user-reported manual label, without counting it as a complete response or a DeepSeek judgment. This is separate from Claude’s second O response on the same question, which is correct. The archive distinction is retained in both the principal scores and the complete-archive sensitivity above.

A.8 SCOPE OF THE CLOSEST-RESOURCE COMPARISON

Our comparison examines task definitions and public assets, not a cross-benchmark ranking. MathVerse’s inspected testmini archive contains 788 distinct problems; 787 text-only prompts exactly match the corresponding Text Dominant prompt. Image removal in that condition generally does not replace the omitted diagram with new text. MathSight supplies original, hand-drawn, photographed, and image-free conditions. Its inspected 661-question file includes algebra, topology, and discrete-mathematics examples as well as its larger calculus component. CombiGraph-Vis examples cover discrete olympiad techniques and include image-bearing jump-counting, coverage, coloring, and tiling problems. GraphVerse’s eleven specified tasks have algorithmic references and process-sensitive evaluation. These comparisons acknowledge overlapping evaluation dimensions and identify the requested mathematical targets and reference assets as the relevant distinctions (Zhang et al., 2025; Wang et al., 2026b; Mahdavi et al., 2025; Sun et al., 2026).

A.9 RECORDED CHRONOLOGY AND REVIEW VISIBILITY

The records separate screening responses from independent evaluation responses, but final quality decisions follow some independent calls. Timestamped screening logs span August 23–September 3, 2026; no corresponding per-request Doubao log was located. Of the 983 complete responses in the current registry, 564 were recorded on September 8–9 and 139 during Hard extension on September 10, 02:21–04:47 UTC. These 703 responses preceded application of the final membership. Reference review, correction, old-answer rejudgment, and the Hard extension overlapped on September 10. The quality-exclusion policy was recorded at 04:47 UTC and membership applied at 04:58 UTC. The remaining 280 Challenge responses were recorded later that day, 15:53–17:47 UTC. The registry assembly date, September 11, is distinct from these response dates.

All 134 reference-review API packets supplied the image and reference but no candidate answer or old-label field. The local correction workflow had broader evidence: independent response texts, model identities, prior judgments, and rejudgment comparisons. The release policy states that independent performance is not an input to the screening rule or quality gate. However, there is no complete access log for the final decision maker, and the gate was not preregistered. These records support separation of answer sets and a documented membership rule; they do not establish global blinding or prospective fixation before independent evaluation. The companion’s timeline preserves the exact timestamps and packet-field audit.

A.10 ARTIFACT AND AUDIT ENTRY POINTS

The accompanying repository linked in Section 1 provides the experimental code, records, and a portable audit companion. Its README identifies the current results and gives the archive extraction and workspace restoration commands. After extraction, the companion opens at `companion/index.html`. Each question page connects the original text and image, current target, content card, reference evidence, four indexed responses per evaluated model and condition, and judgment sources. `questions/index.json` indexes all 62 entries, while `responses/slots.jsonl` indexes 984 distinct positions. The answer directory contains 983 complete response texts. The remaining position contains the explicit manual-label record and no fabricated answer.

`protocol/prompts/` stores all exact expanded solver prompts; `protocol/configurations/` stores the 17 located configurations for these answers. The observed UTC response times and source bindings are retained. `provenance/source_inventory.json` lists copied evidence, including the 121 directly linked reference assets, and `provenance/known_gaps.json` records closure limits. The revision analyses add uniform item exclusion, per-question text-sufficiency arguments, the 12-response case census, and the chronology audit. Paths in the companion resolve within its directory; credentials and machine-specific author paths are omitted.

The companion’s offline verifier checks slot totals, full-answer coverage, links, and packaged-file integrity. It does not rerun every historical mathematical checker or certify every visual reconstruction. Some multi-question checker scripts depend on inputs outside the current 62-question set; source availability is distinguished from a self-contained execution receipt. The repository also preserves earlier screening and independent-response records needed to trace membership and rejudgment.

Large records are distributed in archive parts with a restoration script; credentials and author paths are removed from the release copy while scientific values and labels are retained. Exact reproduction of service calls requires the reader's own credentials and available provider endpoints.

B EXACT TARGETS AND ILLUSTRATIVE RESPONSE CHECKS

B.1 EQUAL GROUP ORDER DOES NOT DETERMINE THE CRITICAL GROUP

For a connected graph with reduced Laplacian $\tilde{L} \in \mathbb{Z}^{(n-1) \times (n-1)}$, the critical group can be represented as

$$K(G) = \mathbb{Z}^{n-1} / \tilde{L}\mathbb{Z}^{n-1}.$$

If the Smith diagonal is $d_1, \dots, d_{n-1}$, then $K(G)$ is the direct sum of the cyclic groups $\mathbb{Z}/d_i\mathbb{Z}$, omitting the factors $d_i = 1$. The product of the factors equals $\det \tilde{L}$, the spanning-tree count (Biggs, 1999; Stanley, 2016; Chaiken, 1982).

For the upper graph in GT-0411, the accepted structure is $K_{2,2,2,2}$. Its reduced Laplacian is

$$\tilde{L} = \begin{pmatrix} 6 & 0 & -1 & -1 & -1 & -1 & -1 \\ 0 & 6 & -1 & -1 & -1 & -1 & -1 \\ -1 & -1 & 6 & 0 & -1 & -1 & -1 \\ -1 & -1 & 0 & 6 & -1 & -1 & -1 \\ -1 & -1 & -1 & -1 & 6 & 0 & -1 \\ -1 & -1 & -1 & -1 & 0 & 6 & -1 \\ -1 & -1 & -1 & -1 & -1 & -1 & 6 \end{pmatrix}.$$

Let Δ_k be the gcd of its $k \times k$ minors, with $\Delta_0 = 1$. Direct integer computation gives

$$(\Delta_0, \dots, \Delta_7) = (1, 1, 1, 1, 3, 36, 1728, 82944).$$

Taking successive quotients produces the Smith diagonal $(1, 1, 1, 3, 12, 48, 48)$. The nontrivial factors are therefore $(3, 12, 48, 48)$. The candidate factors $(6, 6, 48, 48)$ have the same product, but $K/2K$ would have dimension four over $\mathbb{F}_2$, whereas the accepted group has dimension three. This gives a short independently checkable reason why group order alone cannot validate the candidate. The lower graph is a triangular prism, with accepted nontrivial factors $(5, 15)$.

B.2 AN UNSPECIFIED SECTOR COUNT AND A CORRECT CONDITIONAL CALCULATION

The text of GT-0514 defines a central n -gon with one attached triangle per boundary edge and a distinguished central face. Its cell counts are $V = 2n$, $E = 3n$, and $F = n + 1$, so $\chi = 1$ for every compatible n . Its sector action is dihedral, with group order $2n$. The image fixes $n = 8$, giving the accepted complete target

$$\left(16, \frac{a_1^8 + 5a_2^4 + 2a_4^2 + 4a_8 + 4a_1^2a_2^3}{16}, 1 \right).$$

The text does not fix eight rather than another allowed number of sectors. Thus a response can correctly derive $\chi = 1$ and a formula parameterized by n without determining the complete instance-specific target. Conversely, substituting eight without image evidence can produce the accepted tuple. The first Claude text-only response illustrates the latter event; its statement about the rendering is not justified by its input.

B.3 A LOOP CHANGES ONLY PART OF A MULTI-COMPONENT TARGET

GT-0382 asks for the first Betti number, the number of faces on the sphere, and the spanning-tree count of a connected seven-vertex planar multigraph. Its accepted tuple is $(10, 11, 620)$. A GPT first response omits a loop and uses 15 rather than 16 edges. The formulas $b_1 = E - V + 1$ and $F = E - V + 2$ then produce nine and ten. Since a loop belongs to no spanning tree, the spanning-tree count remains 620. This response demonstrates why checking only one component would accept an incomplete solution.

B.4 PROOF OBLIGATIONS CAN DISTINGUISH IDENTICAL TERMINAL VALUES

GT-0019 asks for a hypermap genus and the minimal polynomial of a spectral radius, with a valid derivation and irreducibility argument. The accepted terminal pair is $(5, t^4 - 9t - 36)$. Claude’s first original-image response incorrectly describes the strand permutation as the shift $x_k \mapsto x_{k+3}$ modulo 11. Although its terminal values agree, the wrong instance permutation does not prove the required statement about the depicted object. The complete-answer rubric therefore leaves it incorrect. For this task, a terminal-value-only comparison would not implement the question’s stated requirements.

B.5 A COMPLETE FIRST-RESPONSE CENSUS WITHIN FOUR SELECTED CASES

To compare coarse checks with the specified targets, we read the complete first responses of all three models for GT-0411, GT-0382, and GT-0019 in O and GT-0514 in T. These four cases were selected after their behavior was known. Table 4 records the question-specific projection and complete-target outcomes. The comparison uses the final explicitly corrected answer, not an isolated matching number. In particular, Claude retracts its first GT-0411 factors and ends with $(8, 8, 8, 8, 8, 8)$; the stored parsed-answer field predates that correction. Both versions fail, so the official label is unaffected.

Table 4: Post-hoc census of 12 first responses in four selected cases. Each row covers all three models. A projection keeps only the specified component or terminal values; the complete check follows the original question. These different projections do not define a common alternative benchmark metric. †: one of two concrete sector counts is correct; the third response leaves n unspecified. Complete hits use all three responses.

ID	Input	Projection checked	Projection hits	Complete hits
GT-0411	O	Upper group order, 82944	1/3	0/3
GT-0382	O	Tree count, 620	1/3	0/3
GT-0019	O	Terminal genus–polynomial pair	2/3	1/3
GT-0514	T	Sector count, $n = 8$	1/2†	1/3

GT-0411’s three models all give the correct lower group, but no complete upper group is correct. Gemini alone matches its order. GT-0382’s GPT response gives $(9, 10, 620)$: the tree count matches while the other components fail. For GT-0019, GPT supplies both the correct pair and the required instance derivation and irreducibility proof. Claude’s wrong permutation gives a matrix sharing the true characteristic polynomial, illustrating why that polynomial alone does not establish the depicted instance’s derivation.

In GT-0514 T, GPT leaves n symbolic, Gemini asserts $n = 10$, and Claude guesses $n = 8$. All three derive $\chi = 1$, but only Claude matches the complete fixed target. Two responses report a concrete n , one correct and one wrong; GPT’s unspecified parameter is recorded separately. Both Gemini and Claude attribute their chosen count to a rendering absent from their inputs. The companion records the 12 source bindings, 33 located excerpts, extraction criteria, and exact arithmetic checks. These cases establish inspectable distinctions between target granularity and support; they do not estimate their dataset-wide frequency or calibrate the judge.

C COMPLETE PER-PROBLEM RESULTS

Table 5 gives the four binary outcomes for each question, model, and evaluated input condition. The 42 Challenge-only questions have no T entries in the common analysis. The sum of the displayed positions is 984, including the one explicitly marked manual judgment. A zero denotes an incorrect label, not a missing response, except that the marked position has a user judgment without a complete local answer archive.

Table 5: All four-roll verdict vectors, in roll order (1=Correct, 0=Incorrect). O: original image; T: text only. T was evaluated only on Hard20. †: Claude GT-0560 T roll 2 uses a user-reported Incorrect adjudication without a complete archived response.

ID	Set	O: GPT	O: Gemini	O: Claude	T: GPT	T: Gemini	T: Claude
GT-0004	C	1111	1111	0001	—	—	—
GT-0008	C	1101	0000	1000	—	—	—
GT-0010	C	1111	0111	0000	—	—	—
GT-0012	C	0011	1101	0110	—	—	—
GT-0016	C	1001	1101	0000	—	—	—
GT-0018	C	1011	0000	0000	—	—	—
GT-0019	H	1001	0000	0000	0000	0000	0000
GT-0049	C	1111	0000	0000	—	—	—
GT-0050	H	0010	0000	0000	0000	0000	0000
GT-0061	H	0000	0000	0000	0000	0000	0000
GT-0070	C	1011	0010	0000	—	—	—
GT-0081	C	1010	0000	0000	—	—	—
GT-0084	C	0000	0000	0000	—	—	—
GT-0102	H	0000	0000	0000	0000	0000	0000
GT-0138	C	1100	0011	0000	—	—	—
GT-0163	C	0001	0011	0111	—	—	—
GT-0164	H	1000	0000	0000	0000	0000	0000
GT-0167	C	0010	0000	0000	—	—	—
GT-0168	C	1000	1111	1010	—	—	—
GT-0207	C	1111	1111	0000	—	—	—
GT-0208	C	0111	0000	0000	—	—	—
GT-0216	H	0000	0000	0000	0000	0000	0000
GT-0219	C	1111	0011	0000	—	—	—
GT-0230	C	0111	1111	1100	—	—	—
GT-0239	C	0100	0000	0000	—	—	—
GT-0244	H	0000	1100	0000	0000	0000	0000
GT-0248	H	0000	0000	0000	0000	0000	0000
GT-0255	C	1111	0100	0000	—	—	—
GT-0277	C	1110	1111	0101	—	—	—
GT-0282	C	0010	0000	0000	—	—	—
GT-0288	H	0000	0000	0000	0000	0000	0000
GT-0291	H	0000	0000	0000	0000	0000	0000
GT-0302	H	0000	0000	0000	0000	0000	0000
GT-0305	C	0000	1111	1111	—	—	—
GT-0317	C	1111	0101	0001	—	—	—
GT-0325	C	0010	0000	0000	—	—	—
GT-0361	H	0000	0000	0000	0000	0000	0000
GT-0372	C	0000	0001	0000	—	—	—
GT-0374	C	0000	0000	0000	—	—	—
GT-0382	H	0000	0000	0000	0000	0000	0000
GT-0385	C	0000	0000	0000	—	—	—
GT-0394	C	0001	1011	0000	—	—	—
GT-0395	H	0000	0000	0000	0000	0000	0000
GT-0402	C	0000	1111	0000	—	—	—
GT-0411	H	0000	0000	0000	0000	0000	0000
GT-0424	C	0000	1100	0000	—	—	—
GT-0426	C	0011	0000	0001	—	—	—
GT-0436	C	0110	1111	1100	—	—	—
GT-0437	C	0000	0000	0000	—	—	—
GT-0495	C	0000	1000	0110	—	—	—
GT-0503	C	0110	1001	0000	—	—	—
GT-0514	H	0000	0000	0000	0000	0000	1111
GT-0516	H	0000	0000	0000	0000	0000	0000
GT-0524	C	1000	0000	0000	—	—	—
GT-0526	C	1000	0010	0000	—	—	—
GT-0543	C	0000	0010	0000	—	—	—
GT-0547	C	1001	0000	0000	—	—	—
GT-0554	H	0000	0000	0000	0000	0000	0000

ID	Set	O: GPT	O: Gemini	O: Claude	T: GPT	T: Gemini	T: Claude
GT-0560	H	0000	0000	0100	0000	0000	0000 [†]
GT-0572	C	0000	1011	0000	—	—	—
GT-0573	C	1111	1111	1110	—	—	—
GT-0585	H	0000	0000	0000	0000	0000	0000

D CONTENT AND REFERENCE EVIDENCE INDEX

Table 7 identifies each retained question and its requested target. All entries belong to Challenge62; H marks the nested Hard20 and C its complement. The output form describes the required answer rather than a complete taxonomy of the mathematics. “Group-valued” includes finitely generated abelian groups with a free summand, as in GT-0016 and GT-0018; it does not assert that every group is finite. The accompanying content cards bind these descriptions to the original question and image, the accepted target, and the corresponding verification record. Table 8 lists reference status and a summary of its verification basis; the editable evidence index additionally links the underlying records.

Table 6: Descriptive output forms across the nested evaluation sets. Counts concern questions, not responses.

Output form	Challenge	Hard
Integer tuple	19	6
Symbolic polynomial	14	6
Symbolic tuple	10	3
Integer	8	2
Group-valued	7	2
Structured list	4	1
Total	62	20

Table 7: Content index of all 62 current questions. H denotes Hard20; C denotes Challenge-only. Descriptions follow the saved content inventory.

ID	Set	Output	Requested target
GT-0004	C	Symbolic tuple	Compute alternating-group constituent dimension and character difference.
GT-0008	C	Group-valued	Determine nonunit Smith invariant factors of a reduced graph Laplacian.
GT-0010	C	Symbolic tuple	Compute ribbon-graph and checkerboard transition polynomials; embedding-sensitive graph invariants cross two leaves.
GT-0012	C	Integer tuple	Determine prime exponents of the spanning-tree count of an iterated line graph.
GT-0016	C	Group-valued	Determine free rank and torsion invariant factors of a graph Laplacian cokernel.
GT-0018	C	Group-valued	Determine the invariant-factor decomposition of an integral graph Laplacian cokernel.
GT-0019	H	Symbolic tuple	Compute hypermap genus and minimal polynomial of a block-transition spectral radius; coequal topology and spectral targets.
GT-0049	C	Symbolic polynomial	Determine the characteristic polynomial of a one-period exclusion transition matrix.
GT-0050	H	Integer	Count spanning trees of a line graph.
GT-0061	H	Integer tuple	Count maximal spanning forests, components and cycle-space dimension; forest enumeration is the main nontrivial target.
GT-0070	C	Integer tuple	Compute exponents of three graph critical groups.
GT-0081	C	Integer tuple	Compute graph cycle rank, bridge count and critical-group exponent; exponent is the main algebraic target.
GT-0084	C	Structured list	Count spanning trees of three threshold bipartite graphs.
GT-0102	H	Integer	Count spanning trees of the line graph of a reduced topological graph.

ID	Set	Output	Requested target
GT-0138	C	Integer tuple	Determine rotational symmetry order and reflection-axis count of a planar point set.
GT-0163	C	Symbolic polynomial	Compute reduced integral homology of a sphere quotient collapsing a cubical set.
GT-0164	H	Group-valued	Determine nonunit Smith factors of a spoke-rim line-graph reduced Laplacian.
GT-0167	C	Symbolic tuple	Compute endpoint-permutation cycle type, inversion number and order.
GT-0168	C	Integer	Determine the order of a diagram-derived graph critical group.
GT-0207	C	Integer tuple	Compute sum and product of spanning-tree counts of displayed tree line graphs.
GT-0208	C	Symbolic tuple	Compute cycle-index polynomials for a color-preserving cycle-automorphism group action.
GT-0216	H	Integer tuple	Count faces, internal edges, and side incidences of a planar cell decomposition.
GT-0219	C	Structured list	Recover ramification partitions of a genus one dessin.
GT-0230	C	Group-valued	Determine sandpile group invariant factors.
GT-0239	C	Integer	Count edge distinguished spanning trees of a weighted bundle cycle.
GT-0244	H	Symbolic tuple	Find the least selected word and sum its collection's inversion numbers.
GT-0248	H	Symbolic polynomial	Compute the chromatic polynomial of six disjoint graphs.
GT-0255	C	Symbolic polynomial	Enumerate spanning trees by internal activity.
GT-0277	C	Symbolic tuple	Compute cyclomatic number, directed simple cycle count, and adjacency characteristic polynomial.
GT-0282	C	Integer tuple	Count ribbon graph vertices, bands, and thickened surface boundary components.
GT-0288	H	Structured list	Count displayed closed curves, yellow triangular faces, and distinct fill hues by row.
GT-0291	H	Symbolic polynomial	Compute the adjacency characteristic polynomial of a disjoint clique union.
GT-0302	H	Integer tuple	Compute first Betti numbers of two finite stroke graphs.
GT-0305	C	Symbolic polynomial	Compute a recurrent label multigraph Tutte polynomial.
GT-0317	C	Integer tuple	Count spanning trees and identify the unique articulation vertex.
GT-0325	C	Integer	Count spanning trees of a color domain adjacency graph.
GT-0361	H	Integer tuple	Determine yellow induced subgraph components and vertex counts.
GT-0372	C	Integer tuple	Compute oriented chain boundary strata and coincident seam cancellation.
GT-0374	C	Group-valued	Determine a phase adjacency graph critical group.
GT-0382	H	Integer tuple	Compute spanning tree count alongside graph cycle rank and spherical face count.
GT-0385	C	Structured list	Compute a normalized rational nullspace basis of a weighted skew adjacency matrix.
GT-0394	C	Symbolic polynomial	Compute the weighted perfect matching generating polynomial.
GT-0395	H	Integer tuple	Compute graph cycle rank, complementary reduced component rank, and marked vertex count.
GT-0402	C	Symbolic tuple	Compute spanning tree count and the positive Laplacian spectrum.
GT-0411	H	Group-valued	Determine two critical groups in invariant factor form.
GT-0424	C	Symbolic polynomial	Compute the signed degree enumerator of the distinguished vertex across graph cells.
GT-0426	C	Symbolic tuple	Compute the independence polynomial of the interval inclusion tree and evaluate it at the pictured integer.
GT-0436	C	Symbolic polynomial	Sum matching generating polynomials over the ten displayed graphs.
GT-0437	C	Integer tuple	Count vertices, edges, and complementary faces of an embedded planar mesh.
GT-0495	C	Integer tuple	Compute augmented cycle ranks of three finite core graphs.
GT-0503	C	Symbolic polynomial	Compute the h-polynomial of a noncrossing-diagonal simplicial complex.

ID	Set	Output	Requested target
GT-0514	H	Symbolic tuple	Determine an automorphism-group order, sector-action cycle index, and cell-complex Euler characteristic.
GT-0516	H	Symbolic polynomial	Enumerate marked vertices by their visually assigned height strata.
GT-0524	C	Integer	Compute the Gram determinant of the chamber-set incidence matrix.
GT-0526	C	Integer	Count spanning trees of the central junction-and-trace graph.
GT-0543	C	Integer tuple	Compute rational incidence-matrix rank, with auxiliary blue-sector count.
GT-0547	C	Integer tuple	Count vertices, edges, and triangular simplices of the pictured graph's clique complex.
GT-0554	H	Symbolic polynomial	Compute the spanning-tree enumerator weighted by edges crossing a prescribed vertex partition.
GT-0560	H	Symbolic polynomial	Compute the expanded determinant polynomial of a signed monomial-weighted adjacency matrix.
GT-0572	C	Integer tuple	Determine periodic arrangement length and quotient-graph cycle rank and edge-orbit count.
GT-0573	C	Integer	Count spanning trees of the edge-labeled multigraph, distinguishing parallel tokens.
GT-0585	H	Symbolic polynomial	Construct the polynomial encoding every discrete output transition of a displayed decision-tree classifier.

Table 8: Reference status and verification basis for each current question. S: supported target without answer change; C: corrected target. Summaries describe the recorded checks; they do not imply independent human validation or automatic image verification.

ID	Status	Verification basis
GT-0004	S	Hook-length dimension agrees with a separate corner-removal recurrence.
GT-0008	S	Five boundary hubs give a wheel with 121 spanning trees; 251 minors give nonunit Smith factors (11,11).
GT-0010	S	Exact state-sum calculations support the target.
GT-0012	S	Source reconstruction and exact calculation support the retained target.
GT-0016	S	Contracting the marked endpoints gives a double-edge component, a 4-cycle, and a loop; the integer Smith form is (1,1,2,4,0,0,0).
GT-0018	S	The 57-vertex, 56-edge drawing reduces by 32 unimodular leaf operations to six 4-cycle blocks and a three-vertex path; the cokernel has free rank 7 and six factors of 4.
GT-0019	S	An eleven-arrow reconstruction, cycle and genus calculation, characteristic polynomial, and irreducibility check support the target.
GT-0049	S	Source reconstruction and exact calculation support the retained target.
GT-0050	C	Two image readings recover 32 edges; line-graph cofactors agree with rational elimination.
GT-0061	C	The omitted blue plaquette gives 26 plaquettes, eight components, three cycles, and 146 maximal spanning forests.
GT-0070	S	Source reconstruction and exact calculation support the retained target.
GT-0081	S	The trace has 12 core edges, 11 pendant tails, and 19 vertices. All cofactors give 165; the block critical groups give exponent 165.
GT-0084	S	Source reconstruction and exact calculation support the retained target.
GT-0102	C	Apparent crossings resolve into distinct junctions. The tree has 14 trivalent vertices and 16 leaves; line-graph cofactor and block product agree.
GT-0138	C	All 210 marker centers match the triangular-lattice annulus. Six rotations and six reflections preserve it; the outermost six-point layer excludes additional symmetries.
GT-0163	S	The red cubical set has two components, each with Euler characteristic 1 and no first homology, supporting the quotient polynomial.
GT-0164	S	Three qualifying spokes determine a complete graph on four vertices; 251 minors of its line-graph reduced Laplacian give factors (2,8,24).
GT-0167	S	Source reconstruction and exact calculation support the retained target.
GT-0168	S	An upper block with 30 trees joins a 4-cycle at an articulation; cofactor and enumeration of 495 candidate edge sets both give 120.

ID	Status	Verification basis
GT-0207	S	Source reconstruction and exact calculation support the retained target.
GT-0208	S	Source reconstruction and exact calculation support the retained target.
GT-0216	C	Two image computations find 102 blue sites, 102 bounded cells, and nine boundary segments per side; Euler and degree counts give (102,271,578).
GT-0219	S	Source reconstruction and exact calculation support the retained target.
GT-0230	S	Image reading and exact Smith normal form support the target.
GT-0239	C	Separate assistant source reconstructions and exact computations agree.
GT-0244	C	Eight blue dashed edges define the words; inversion enumeration preserves leading zeros and sums to 31.
GT-0248	S	The six diagrams are three 3-stars, two 4-cycles, and an 8-cycle. Edge-subset polynomial calculations agree with direct color enumeration.
GT-0255	S	Source reconstruction and exact calculation support the retained target.
GT-0277	S	Source reconstruction and exact calculation support the retained target.
GT-0282	S	The evidence records a source-specific mathematical or rule-based resolution.
GT-0288	C	Separate assistant source reconstructions and exact computations agree.
GT-0291	S	Pixel interpolation recovers all 13 bar heights; the disjoint-clique spectrum matches the complete target.
GT-0302	C	Visible-stroke and exterior-gap rules prohibit hidden bridges, giving zero pockets on the left and five on the right.
GT-0305	C	Exactly twice-occurring labels form a 6-cycle; labels occurring three times are excluded. Two image readings and all 64 edge-subset calculations agree.
GT-0317	C	The image agrees with the edge list; fourteen cofactors and tree enumeration correct the count to 3909.
GT-0325	S	The evidence records a source-specific mathematical or rule-based resolution.
GT-0361	C	The image has 64 yellow vertices. A 63-edge spanning-tree witness connects all four groups; full edge enumeration is not required.
GT-0372	C	The evidence records a source-specific mathematical or rule-based resolution.
GT-0374	S	A connected white corridor separates red and black. The region graph has two triangles sharing a vertex, with Smith factors (3,3).
GT-0382	C	Seven vertices and 16 edges, including the repeated arc and loop, give (10,11,620); all seven cofactors agree with tree enumeration.
GT-0385	C	Two image readings confirm the directed edge from 3 to 6. Exact rank is 2; the requested first two free-column basis vectors are well defined within the four-dimensional kernel.
GT-0394	S	Source reconstruction and exact calculation support the retained target.
GT-0395	C	Separate assistant image inspections and exact computation support the target.
GT-0402	S	The eleven-edge graph has six explicit orthogonal Laplacian eigenvectors with eigenvalues (0,3,3,5,5,6); all cofactors and enumeration of 462 candidate trees give 225.
GT-0411	C	Determinantal divisors give lower-prism factors (5,15), with 75 trees; the upper-graph factors are also supported.
GT-0424	C	The 21 signed cells and their degrees at vertex 6 sum to the corrected polynomial.
GT-0426	S	The evidence records a source-specific mathematical or rule-based resolution.
GT-0436	S	Source reconstruction and exact calculation support the retained target.
GT-0437	C	Five black tracks occur in each family; the prescribed count ignores colored curves.
GT-0495	S	Source reconstruction is documented in the reference evidence.
GT-0503	S	A separate source reconstruction and exact calculation support the target.
GT-0514	C	Separate assistant image inspections and exact computation support the target; the cell and symmetry formulas are detailed in Appendix B.2.
GT-0516	C	The five strata contain 1, 14, 24, 12, and 1 explicit node glyphs; multiple pixel thresholds and whole-glyph grouping agree.
GT-0524	S	Eighteen orange markers have matching colors. Row-set sizes (5,7,5,1) and their intersections give Gram determinant 93.
GT-0526	C	Twelve vertices and twenty edges give 24,000 spanning trees by all cofactors and edge-subset enumeration.
GT-0543	C	Seven clusters of eight glyphs give 56 vertices and 84 edges; the exact incidence rank is 55.
GT-0547	C	Separate assistant image inspections and exact computation support the target.
GT-0554	C	The corrected edge set removes two absent edges; exact weighted cofactors and full multiedge-tree enumeration agree.

ID	Status	Verification basis
GT-0560	C	Twelve arrows include the red arrow from 3 to 4 and no red loop at 3. Three determinant methods agree; the directional wording erratum is recorded in Appendix A.7.
GT-0572	S	A separate source reconstruction and exact calculation support the target.
GT-0573	S	A separate source reconstruction and exact calculation support the target.
GT-0585	C	All 48 paths identify no jump at 13 and 27; fifteen active boundary roots include four missing from the original reference.

E REFERENCE-CORRECTION SENSITIVITY

The fixed-response comparison in Table 9 measures the effect of reference correction and rejudgment on identical answer texts and fixed question memberships. The 53/136-question archived analysis sets, and the corresponding 52/134-question subsets without the earliest two corrected references, are separate from the quality-filtered Challenge62 and Hard20 membership reported in the main text. These sets include questions later excluded for unresolved quality or reference issues. The comparison therefore documents label sensitivity; it is not an additional current benchmark leaderboard.

Table 9: Scores on fixed archived answer sets before and after reference correction and rejudgment. Rates are percentages and changes are percentage points. Answer texts and denominators remain unchanged within a row. Membership is held fixed for this sensitivity analysis.

Fixed membership	Model	Input	Slots	Saved	Corrected	Δ (pp)
Hard53	GPT	O	212	6.60	33.49	+26.89
Hard53	GPT	T	212	2.36	4.25	+1.89
Hard53	Gemini	O	212	2.36	34.91	+32.55
Hard53	Gemini	T	212	5.66	8.02	+2.36
Hard53	Claude	O	212	4.25	11.79	+7.55
Hard53	Claude	T	212	6.13	4.25	-1.89
Challenge136	Claude	O	544	11.40	16.54	+5.15
Hard52	GPT	O	208	5.29	32.69	+27.40
Hard52	GPT	T	208	0.48	2.40	+1.92
Hard52	Gemini	O	208	0.96	34.13	+33.17
Hard52	Gemini	T	208	3.85	6.25	+2.40
Hard52	Claude	O	208	2.40	10.10	+7.69
Hard52	Claude	T	208	4.33	2.40	-1.92
Challenge134	Claude	O	536	10.26	15.49	+5.22

F ADDITIONAL REPEATED-SUCCESS AND ABLATION METRICS

The complete Hard20 ablation metrics and question-level success distributions are given below. These reuse the same positions as the main results and add no scoring positions. Percentages are rounded only for presentation.

Table 10: Hard20 scores under O and T. Rates and intervals are percentages. Correct counts refer to 80 response positions per model and condition; Acc@1, Pass@4, and Stable@4 use 20 questions.

Model	Input	Correct	Acc@1	MeanAcc	Pass@4	Stable@4
GPT-5.6 Luna	O	4/80	10.00 [2.8, 30.1]	5.00 [0.0, 11.2]	15.00 [5.2, 36.0]	0.00 [0.0, 16.1]
GPT-5.6 Luna	T	0/80	0.00 [0.0, 16.1]	0.00 [0.0, 0.0]	0.00 [0.0, 16.1]	0.00 [0.0, 16.1]
Gemini 3.6 Flash	O	2/80	5.00 [0.9, 23.6]	2.50 [0.0, 7.5]	5.00 [0.9, 23.6]	0.00 [0.0, 16.1]
Gemini 3.6 Flash	T	0/80	0.00 [0.0, 16.1]	0.00 [0.0, 0.0]	0.00 [0.0, 16.1]	0.00 [0.0, 16.1]
Claude Opus 4.8	O	1/80	0.00 [0.0, 16.1]	1.25 [0.0, 3.8]	5.00 [0.9, 23.6]	0.00 [0.0, 16.1]
Claude Opus 4.8	T	4/80	5.00 [0.9, 23.6]	5.00 [0.0, 15.0]	5.00 [0.9, 23.6]	5.00 [0.9, 23.6]

Table 11: Question counts by number of successful responses among four attempts. Challenge62 includes Hard20; the two rows for a model must not be added.

Subset	Model	0	1	2	3	4	Recovered after roll 1
Challenge62	GPT	28	11	9	6	8	13
	Gemini	35	6	7	5	9	10
	Claude	48	5	6	2	1	8
Hard20	GPT	17	2	1	0	0	1
	Gemini	19	0	1	0	0	0
	Claude	19	1	0	0	0	1
Challenge-only42	GPT	11	9	8	6	8	12
	Gemini	16	6	6	5	9	10
	Claude	29	4	6	2	1	7

G UNIFORM EXCLUSION OF THE CONFLICTING ITEM

This sensitivity removes GT-0560 for every model and condition, addressing its contradictory directional wording and the missing complete Claude T response on a common population. “Challenge61” and “Hard19” below denote this exclusion analysis only; the declared benchmark memberships remain Challenge62 and Hard20. The retained positions are all complete: $61 \times 3 \times 4 + 19 \times 3 \times 4 = 960$, comprising 180 correct and 780 incorrect labels. The reduction of 24 positions removes one correct response and 23 incorrect positions, including the manual decision. No remaining response, reference, or label is changed.

The O success union changes from 43/62 to 42/61 and from 5/20 to 4/19. Claude’s only Hard O success is removed; its four T successes on GT-0514 remain. All three Hard19 paired T-minus-O intervals still include zero, and the all-zero O or T means have degenerate empirical bootstrap intervals. This analysis shows the dependence of individual scores on the item while preserving the broader pattern of low Hard success. It does not determine how any model would respond to a corrected prompt.

Behavior sensitivity removes all six GT-0560 first responses without recoding the other annotations. Among 57 O responses, 49 have an instance-statement conflict, two a modeling conflict, one a condition conflict, one a computational conflict, one an unresolved error, and three consistent target and checked reasoning. Among 57 T responses, 22 are justified refusals, four general methods, nine final guesses, two empty-object interpretations, and 20 unsupported instance answers. Explicit absent-image attribution occurs in eight, with six ambiguous and 43 negative. These are the original model-assisted categories on a reduced population, not a new independent annotation study.

Table 12: Uniform GT-0560 exclusion. All rates and intervals are percentages. MeanAcc intervals resample questions; binary metric intervals are Wilson intervals. C61 and H19 denote Challenge61 and Hard19. Each row retains four responses per question.

Set	Model	Input	Correct	Acc@1	MeanAcc	Pass@4	Stable@4
C61	GPT	O	79/244	34.43 [23.7, 47.0]	32.38 [23.8, 41.4]	55.74 [43.3, 67.5]	13.11 [6.8, 23.8]
C61	Gemini	O	71/244	27.87 [18.2, 40.2]	29.10 [19.7, 38.9]	44.26 [32.5, 56.7]	14.75 [8.0, 25.7]
C61	Claude	O	26/244	9.84 [4.6, 19.8]	10.66 [5.3, 16.8]	21.31 [12.9, 33.1]	1.64 [0.3, 8.7]
H19	GPT	O	4/76	10.53 [2.9, 31.4]	5.26 [0.0, 11.8]	15.79 [5.5, 37.6]	0.00 [0.0, 16.8]
H19	Gemini	O	2/76	5.26 [0.9, 24.6]	2.63 [0.0, 7.9]	5.26 [0.9, 24.6]	0.00 [0.0, 16.8]
H19	Claude	O	0/76	0.00 [0.0, 16.8]	0.00 [0.0, 0.0]	0.00 [0.0, 16.8]	0.00 [0.0, 16.8]
H19	GPT	T	0/76	0.00 [0.0, 16.8]	0.00 [0.0, 0.0]	0.00 [0.0, 16.8]	0.00 [0.0, 16.8]
H19	Gemini	T	0/76	0.00 [0.0, 16.8]	0.00 [0.0, 0.0]	0.00 [0.0, 16.8]	0.00 [0.0, 16.8]
H19	Claude	T	4/76	5.26 [0.9, 24.6]	5.26 [0.0, 15.8]	5.26 [0.9, 24.6]	5.26 [0.9, 24.6]

Table 13: Paired MeanAcc differences after uniform exclusion, in percentage points. All pairs use the same retained questions on both sides, 100,000 bootstrap replicates, and seed 20260910.

Population	Comparison	Difference	95% CI
Challenge61	GPT O minus Gemini O	+3.28	[-6.97, 13.52]
Challenge61	GPT O minus Claude O	+21.72	[12.30, 31.56]
Challenge61	Gemini O minus Claude O	+18.44	[11.07, 26.23]
Hard19	GPT O minus Gemini O	+2.63	[-5.26, 10.53]
Hard19	GPT O minus Claude O	+5.26	[0.00, 11.84]
Hard19	Gemini O minus Claude O	+2.63	[0.00, 7.89]
Hard19	GPT T minus GPT O	-5.26	[-11.84, 0.00]
Hard19	Gemini T minus Gemini O	-2.63	[-7.89, 0.00]
Hard19	Claude T minus Claude O	+5.26	[0.00, 15.79]

H PER-QUESTION TEXT-SUFFICIENCY ARGUMENTS

Table 14 records the omitted facts, compatible alternatives or local changes, and affected target components for Hard20. A change to one required component leaves the complete target undetermined, even when other components follow from text. These model-assisted arguments range from explicit small alternatives to changes that hold other structure fixed; they are not 20 complete replacement images or independently certified structures.

Three qualifications remain. GT-0216 lacks explicit alternative cell complexes satisfying every degree condition. GT-0554 does not specify the added edge’s endpoints or expanded polynomial. GT-0585 changes the active boundary, but two new factors would require an interior singleton. The all-20 audit finding retains these evidential limits. Original annotations, source lines, and scope notes accompany the table in the companion.

Table 14: Existing Hard20 text-sufficiency arguments. Alternatives describe compatible changes to facts omitted from the text; they are not a collection of complete replacement images. The qualifications below and in the text delimit the evidence.

ID and missing fact	Existing alternative or local change	Changed target and qualification
GT-0019 Endpoint permutation.	On eleven darts, $\sigma = r$ versus $\sigma = r^{-1}$.	Genus: 5 versus 0.

ID and missing fact	Existing alternative or local change	Changed target and qualification
GT-0050 Marker count and adjacency.	Use $G=K_3$ versus $G=K_2$.	Line-graph tree count: 3 versus 1.
GT-0061 Filled torus plaquettes.	On $\mathbb{Z}_{10} \times \mathbb{Z}_{10}$, use $\{(0,0)\}$ versus $\{(0,0),(2,2)\}$.	Tuple: (1,0,1) versus (2,0,1).
GT-0102 Stroke branches and endpoints.	After the prescribed deletion/suppression, retain a path versus a tree with one degree-three branch.	Line-graph tree count: 1 versus 3.
GT-0164 Spoke count N.	Use the specified construction with $N=3$ versus $N=4$.	Nonunit Smith factors: (2,8,24) versus (33,264).
GT-0216 Cell count N; boundary count b.	A generic cell complex versus a local generic refinement increasing N.	N changes; $E = 3N + 1$ and $E_{\text{int}} = 3N + 1 - b$. Complete compatible cell pair not supplied.
GT-0244 Dashed edges; fills and labels.	Relabel the unfilled endpoint of one dashed edge from 01325 to 13025.	Tuple: (1,01325,1) versus (1,13025,3).
GT-0248 Six source adjacencies.	Replace one selected P_4 by C_4 ; hold the other five graphs fixed.	Chromatic factor: $q(q-1)^3$ versus $(q-1)^4 + (q-1)$.
GT-0288 Panel curves, faces and hues.	Change the upper-right face colors from yellow/green to yellow/yellow.	Distinct-hue component p1: 2 versus 1.
GT-0291 Thirteen bar heights.	Change one bar from height 6 to 8, away from half-integers; hold other bars fixed.	One characteristic-polynomial factor changes from K_6 to K_8 .
GT-0302 Open versus closed returns.	Close a genuine gap in one return; hold the rest fixed.	Local cycle contribution: 0 versus 1; a Betti component changes.
GT-0361 Yellow nodes and adjacency.	Two yellow nodes with versus without an edge.	Tuple: (1,2,2) versus (2,1,2).
GT-0382 Arc endpoints, loops and multiplicities.	Add a loop at f to a compatible seven-vertex plane multigraph, preserving the other arcs.	Cycle rank and face count each increase by 1; tree count is unchanged.
GT-0395 Incidences and marked vertices.	Add one degree-two arrowhead mark on an existing arc.	Arrowhead count a increases by 1; r and s are unchanged.
GT-0411 Two graph adjacencies.	Use C_4 versus K_4 for the upper graph; hold the lower graph fixed.	Upper critical group: $\mathbb{Z}/4\mathbb{Z}$ versus $(\mathbb{Z}/4\mathbb{Z})^2$.
GT-0514 Sector count n.	Use the central polygon and triangular fan with $n=6$ versus $n=8$.	Group order: 12 versus 16; Euler characteristic remains 1.
GT-0516 Intermediate-stratum node counts.	Add one visible node mark at height 1, preserving the five strata.	Coefficient of t increases by 1.
GT-0554 Source edge lists.	Take $A=B=P_6$; add a cross-I/J edge in one source, preserving the stated matching.	Weighted tree polynomial changes. Added edge and expansion unspecified.
GT-0560 Arrow endpoints and labels.	Change the exponent pair (r,s) on one loop; hold other arrow data fixed.	Trace of M changes, hence the coefficient of z in $\det(I - zM)$.
GT-0585 Thresholds and terminal classes.	From all leaves labeled no jump, relabel one reachable singleton as jump.	Active boundary and transition polynomial change. Two factors require an interior singleton.